

***MICHIGAN DEPARTMENT OF NATURAL RESOURCES & ENVIRONMENT
RESOURCE MANAGEMENT-WATER RESOURCES DIVISION
POINT SOURCE MONITORING***

Report of a
Municipal Wastewater Survey
Conducted at
Cadillac WWTP
NPDES No. MI0020257
Wexford County
Cadillac, MI
January 24, 2011

SURVEY SUMMARY

Wastewater monitoring was performed during one twenty-four hour survey period starting Monday, January 24, 2011.

The results of this survey are compared to the final limitations in the facility's National Pollutant Discharge Elimination System (NPDES) Permit, No. MI0020257, in Table 4.

The results of this survey are compared to the results reported by the facility in their January Discharge Monitoring Report (DMR) in Table 4.

A composite sample from monitoring point (001A) was split with the facility to compare laboratory analyses. The results are shown in Table 5 as 001A-F.

The results of this survey are compared to the results of the previous survey performed in September, 2009 in Table 6.

The results of the physical, chemical and bacteriological analyses are presented in Tables 1 & 2.

There is no Table 3 in this report.

SURVEY PROCEDURE

The flow and samples were obtained as follows:

Sample Location	Flow Measurement	Sampling Methods
001A samples collected from plant's automatic composite sampler post UV disinfection, excepting fecal coliform, pH, Temp., and D.O. which were sampled from pipe, post UV just prior to discharge.	Totalizer reading.	Grab composite and individual grab samples.
Upstream samples collected from the Clam River @ Smith St. bridge. (N44.26330,W085.40099)	-----	Individual grab samples.

An automatic sampler composites samples at timed intervals. Samples may be proportional to the instantaneous flow over the weir or through the flume.

A grab composite consists of a series of individual grabs composited into one sample.

An individual grab is a single instantaneous sample.

Samples were analyzed by the MDNRE Laboratory located in Lansing and TriMatrix Labs of Grand Rapi

Samples were preserved according to Table 7. The results of the physical, chemical, and bacteriological analyses are presented in Tables 1 & 2. Letter codes for laboratory results are defined in Table 7.

There is no Table 3 in this report.

TABLE 1 - Analyses of Composite Samples

Monitoring Point

Survey Period From:

To:

Computed Flow Rate (MGD):

001A

1/24/2011 10:30

1/25/2011 10:30

2.1

	<u>mg/L</u>	<u>lbs/day</u>
Suspended Solids	4	70
CBOD-5	1.7 J	30
NH3	<0.1 ND,D	NA
Total Phosphorus	0.13	2.3
Alkalinity (as CaCO3)	110	NA
Hardness (as CaCO3)	257	NA

	<u>ug/L</u>	<u>lbs/day</u>
Total Zinc (Zn)	69	1.2
Total Selenium (Se)	<1 ND	NA

TABLE 2 - Analyses of Grab Samples

Monitoring Point	001A	001A	001A	Clam River Upstream
Matrix	Water	Water	Water	Water
Date Collected	1/24/2011	1/24/2011	1/25/2011	1/24/2011
Time Collected	10:40	18:40	9:40	11:20
	<u>mg/L</u>	<u>mg/L</u>	<u>mg/L</u>	<u>mg/L</u>
Temperature (F)	52	51	52	40
pH (S.U.)	7.2	7.1	7.3	7.7
Dissolved Oxygen	10.4	10.4	10.1	---
Available Cyanide	0.006	---	---	---
Alkalinity (as CaCO3)	---	---	---	79
Hardness (as CaCO3)	---	---	---	96
Fecal Coliform (cts/100 ml)	530 H	740 H	1300	---
	<u>ng/L</u>	<u>ng/L</u>	<u>ng/L</u>	<u>ng/L</u>
Total Mercury	0.8	---	---	---

TABLE 4
Comparison of the Survey Results with the Facility's NPDES Permit and Discharge Monitoring Report

Parameter (unit)	Final Effluent Limitations				January Discharge Monitoring Report							Survey Results		
	Maximum Monthly	Maximum 7-day	Minimum Daily	Maximum Daily		Average	Maximum 7-day	Minimum Daily	Maximum Daily	1/24/2011	1/25/2011		Composite	(grab)
Monitoring Point (001A)														
Flow (MGD)	(report)			(report)		1.981			2.205	2.023	2.028		2.1	
CBOD-5 (mg/L)	25			34		2.657			3.400	3.4	2.6		1.7 J	
CBOD-5 (lbs/day)	670	910				44.673	45.592			57.371	43.969		30	
Suspended Solids (mg/L)	30	45				3.1	3.1			3.4	3.2		4	
Suspended Solids (lbs/day)	800	1200				51.7	52.8			57.371	54.115		70	
Suspended Solids % removal			85					97.683		97.763	98.75			
Ammonia (mg/L)	4.2					0.1				0.05	0.08		<0.1 ND,D	
Ammonia (lbs/day)	110					1.3				0.844	1.353		NA	
Total Phosphorus (mg/L)	0.5					0.14				0.15	0.12		0.13	
Total Phosphorus (lbs/day)	13					2.425				2.531	2.029		2.3	
Fecal Coliform (cts/100 ml)	200	400				75.524	116.2			78	200			(530 H, 740 H, 1300)
pH (S.U.)			6.5	9.0				6.8	7.5	6.96	7.11			(7.2 , 7.1 , 7.3)
Dissolved Oxygen (mg/L)			5.0					9.1		10.5	10.2			(10.4 , 10.4 , 10.1)

Table 5
Comparison of the Laboratory Analytical Results
Obtained by the Facility and the MDNRE Lab

Monitoring Point

Sample Type

Date Collected

Time Collected

001A-F

Composite

1/25/2011

7:00

	Facility	MDNRE
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	3.2	5
CBOD-5	2.6	2.1
NH3	0.0836	<0.1 ND ₃ D
Total Phosphorus	0.120	0.15

Table 6 - Comparison of Results with Previous Survey

Monitoring Point		001A
Matrix		Water
Survey Period From:	09/08/09	01/24/11
To:	09/09/09	01/25/11
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	4	4
CBOD-5	2	1.7 J
NH3	< 0.2 ND,D	<0.1 ND,D
Total Phosphorus	0.27	0.13
Alkalinity (as CaCO3)	83	110
Hardness (as CaCO3)	206	257
	<u>ug/L</u>	<u>ug/L</u>
Total Zinc (Zn)	18	69
Total Selenium (Se)	<1 ND	<1 ND

TABLE 7 - Sample Preservation

Parameter	Preservative
COD/TOC/Phenol/Nutrients (Chlorine Absent)	5 drops conc. H ₂ SO ₄ /250 ml (to pH <2)
Available Cyanide	Treated with Lead Carbonate, filtered and preserved with Sodium Hydroxide to a minimum pH of 12.
Dissolved Oxygen (D.O.)	Meter reading.
Total Metals	2 ml 1:1 HNO ₃ /250 ml (to pH <2).
Microbiology	2 drops 10% sodium thiosulfate/125 ml to dechlorinate sample.
Samples preserved as required, cooled to 4 degrees Centigrade with chain of custody maintained.	

Lab Letter Codes

D - Analyte value quantified from a dilution(s); reporting limit (RL) raised.

H - Recommended laboratory holding time was exceeded.

J - Result was detected between MDL and RL and is considered estimated.

ND - Not detected.

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Laboratory Analyses by: Michigan Department of Natural Resources & Environment Laboratory
TriMatrix Labs of Grand Rapids, MI

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